CYBERJOAR AI | OC.41335.2026.59218

Assignment Submission Report

Multi-Domain AI Dashboard Development · Problem Statements 1, 3 & 4

Submitted By

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Problem Statements	Live Deployments	Public Access	GitHub Repository
3 / 5	4	100%	1

Live Deployment URLs

All four applications are publicly deployed on Vercel. No login, installation, or API key is required — open in any modern browser (Chrome / Firefox / Edge recommended).

Application	Public URL
Landing Page	https://pgk-placement-drive.vercel.app
PS1 — Intelligence	https://pgk-ps1-intelligence.vercel.app
PS3 — Urban Growth	https://pgk-ps3-urban-growth.vercel.app
PS4 — Traffic Intel	https://pgk-ps4-traffic-intel.vercel.app

Executive Summary

This report documents the design, architecture, and logic of three interactive AI-powered web dashboards developed as part of the CyberJoar AI placement-drive assignment. Each application addresses a specific problem statement from the brief, delivering fully functional, production-grade interfaces accessible via public Vercel deployments.

The three implemented solutions are:

- PS1 — STRATFUSE: Multi-Source Intelligence Fusion Dashboard combining OSINT, HUMINT, and IMINT data on a unified geospatial interface
- PS3 — URBANVECTOR: Predictive Urban Growth Analytics Engine for real estate investment decision-making using a Growth Velocity Score (GVS) algorithm
- PS4 — TRAFFIQ: Hybrid Predictive Traffic Congestion Model fusing CCTV computer-vision simulation with LSTM-based congestion forecasting

All applications are single-file HTML implementations requiring zero server setup, built with Vanilla JavaScript, Leaflet.js for geospatial rendering, and Chart.js for data visualization. The codebase is fully extensible — simulated data sources can be replaced with live API connections without architectural changes.

PROBLEM STATEMENT 01

STRATFUSE — Multi-Source Intelligence Fusion Dashboard

1.1 Application Overview

STRATFUSE addresses the core operational challenge of fragmented intelligence: analysts toggling between database viewers, spreadsheet tools, and image galleries with no unified "common operating picture." The dashboard consolidates OSINT, HUMINT, and IMINT data onto a single interactive Leaflet terrain map with real-time ingestion, filtering, and hover-activated imagery pop-ups.

Stack Component	Technology Used	Purpose	Replaces (Production)
Map Engine	Leaflet.js 1.9.4	Geospatial tile rendering	Mapbox / OpenLayers
Tile Provider	CartoDB Dark Matter	Terrain-style dark base map	Mapbox Satellite / ArcGIS
File Parsing	HTML5 File API	CSV / JSON / JPG ingestion	Apache Kafka / S3 SDK
Imagery Display	Leaflet Popup + base64	Hover IMINT image render	AWS S3 pre-signed URLs
Frontend	Vanilla JS / CSS3	UI, state management	React + Redux

1.2 Core Logic & Architecture

Unified Node Schema

All ingestion pathways normalize their output into a single node schema, ensuring the map renderer and inspector panel are completely source-agnostic:

```
{ id, type, lat, lng, title, summary, description, source, confidence,
timestamp, imageUrl }
```

Multi-Modal Ingestion Pipeline

Four ingestion pathways are implemented:

1. **MongoDB Connector** — Simulates async document fetch from an OSINT collection. In production, replaces with a MongoDB Atlas Data API call via fetch().
2. **AWS S3 Sync** — Fetches IMINT assets with associated base64 imagery metadata. Replaceable with the AWS SDK v3 GetObject call.
3. **HUMINT JSON Parser** — Parses field reports in JSON format. Supports nested objects with field-mapping fallbacks.
4. **Drag-and-Drop File API** — HTML5 ondrop/onchange handlers route to JPG (FileReader.readAsDataURL), JSON (JSON.parse), or CSV (line-split + header extraction) processors.

Hover-and-View Interactivity

Leaflet markers bind popups on the mouseover event (not just click), satisfying the hover-activated requirement. IMINT nodes populate an <img> tag with the node's imageUrl property, rendering satellite/field imagery instantly within the popup without navigation.

Confidence Scoring

Each node carries a 0–100 confidence score rendered as a progress bar in the Node Inspector panel, color-mapped by intelligence type (OSINT=cyan, HUMINT=green, IMINT=orange). Filter buttons toggle Leaflet layer visibility per source type.

1.3 Functional Requirements Coverage

Requirement	Implementation	Status
MongoDB & S3 Ingestion	Simulated async connectors with real node schema output	✓ **Implemented**
Manual CSV/JSON Upload	HTML5 File API drag-and-drop with field-mapped parsers	✓ **Implemented**
IMINT JPG Upload	FileReader.readAsDataURL renders image in popup and inspector	✓ **Implemented**
Terrain Map Interface	Leaflet + CartoDB Dark Matter tiles (Mercator projection)	✓ **Implemented**
Geospatial Node Dots	Color-coded DivIcon markers plotted at lat/lng coordinates	✓ **Implemented**
Hover-and-View Popup	mouseover event triggers Leaflet popup with inline IMINT image	✓ **Implemented**

PROBLEM STATEMENT 03

URBANVECTOR — Predictive Urban Growth Analytics Engine

2.1 Application Overview

URBANVECTOR solves the "insight gap" in real estate investment: the inability to correlate government infrastructure declarations with commercial market signals to identify future high-growth zones. The engine aggregates four weighted data streams into a composite Growth Velocity Score (GVS) and projects results onto an interactive heatmap covering Delhi NCR, Mumbai MMR, Bengaluru, and Pune.

Data Stream	Source Modeled	Analytical Role	Default Weight
Municipal Declarations	Govt. portals, CLU filings	Lead indicator — future supply	**40%**
Listing Density	99acres / MagicBricks	Demand saturation indicator	**25%**
Price Velocity	Real estate portals (YoY ₹/sqft)	Appreciation momentum	**20%**
Rental Absorption	Online rental portals	Real residency demand signal	**15%**

2.2 GVS Algorithm

The Growth Velocity Score is computed as a user-adjustable weighted linear combination of the four normalized data streams:

```
GVS = (w1 × MunicipalScore) + (w2 × ListingDensity) + (w3 × PriceVelocity) + (w4 × RentalAbsorption)
```

Where $w_1 + w_2 + w_3 + w_4$ sum to 100. The municipal declaration weight defaults to 40% because infrastructure announcements function as the strongest lead indicator of future capital appreciation — a causal factor, not merely correlated.

Tiered Classification

Tier	GVS Range	Map Color	Investment Signal
HIGH GROWTH	80 – 100	● Green (#22c55e)	Strong buy signal; 12-24 month horizon
MODERATE	60 – 79	● Amber (#f59e0b)	Watch-and-enter; 30-48 month horizon

LOW	0 – 59	● Red (#ef4444)	Avoid or monitor; 60+ month horizon

2.3 60-Month Forecast Model

The GVS forecast chart uses an exponential saturation model that mimics the realistic S-curve of real estate appreciation cycles:

```
GVS(t) = GVS₀ + (100 − GVS₀) × (1 − e^(−0.08t))
```

This models the diminishing-returns growth pattern: early-stage appreciation is steep, flattening as the zone matures and prices approach market equilibrium. The model is re-evaluated when the user adjusts weight sliders and clicks "Run Analysis."

2.4 City Coverage

- Delhi NCR — 6 micro-markets (Dwarka Expressway, Noida Sector 150, L-Zone, New Gurugram, Greater Faridabad, Kundli-Manesar)
- Mumbai MMR — 4 micro-markets (Navi Mumbai / NMIA zone, Thane-Kalyan, Panvel-Khopoli, Virar-Vasai)
- Bengaluru — 4 micro-markets (Devanahalli / BIAL, Whitefield, Sarjapur Road, Electronic City Ph-II)
- Pune — 3 micro-markets (Hinjewadi Ph-III, Wagholi-Kesnand, Undri-Pisoli)

PROBLEM STATEMENT 04

TRAFFIQ — Hybrid Predictive Traffic Congestion Intelligence

3.1 Application Overview

TRAFFIQ addresses the "reactive trap" in urban traffic management: systems that detect jams only after they form. The dashboard fuses two data modalities — CCTV-based computer vision (micro-view) and GPS/Google Maps data (macro-view) — with a temporal LSTM prediction layer to forecast standstill congestion up to 45 minutes ahead.

CCTV Nodes (Delhi)	Forecast Horizons	Live Tick Rate	Max Jam Probability
8	3	3s	L5

3.2 System Architecture

Module	Technology	Function
Computer Vision Layer	HTML5 Canvas + YOLO bbox simulation	Renders animated CCTV feed; density-tied vehicle speed; overlays detection boxes at congestion Level 3+
Geospatial Layer	Leaflet.js + CircleMarker + DashArray rings	Color-coded node markers; dashed prediction rings for LSTM-flagged zones; clickable popups with full telemetry
LSTM Prediction Engine	Chart.js multi-line time-series	Plots 45-minute congestion probability curves for top 3 corridors; confidence degrades with forecast horizon (91%/84%/72%)
Cross-Modal Fusion	Alert classification engine	Detects mismatch between Google Maps baseline and CCTV vehicle counts; filters false positives; classifies as CRITICAL / WARNING / LSTM PRED
Live Data Simulation	setInterval ticker (3s)	Stochastic drift model updates density/speed/vehicle counts across all 8 nodes; Chart.js updates without re-render lag

3.3 CCTV Simulation Logic

The HTML5 Canvas module renders a real-time traffic feed simulation. Eight car sprites traverse horizontal lanes; their speed is dynamically computed from the selected node's density index:

```
vehicleSpeed = baseSpeed × (1 − densityRatio × 0.85)
```

At congestion Level 3 or above, the canvas overlays YOLO-style bounding boxes with "VEH" classification labels — simulating what a production YOLO-v8 or Faster R-CNN output would produce when processing IP camera frames at 15-30 FPS. The live density progress bar at the top of the feed updates in real time.

3.4 Congestion Level Classification

Level	Label	Color	Trigger Condition
L1	FREE FLOW	● Teal	Density <40%; Avg speed >50 km/h
L2	SLOW	● Amber	Density 40-65%; Speed 25-50 km/h
L3	CONGESTED	● Red	Density 65-85%; Speed 10-25 km/h; YOLO bboxes active
L4	**STANDSTILL**	● Critical Pink	Density >85%; Speed <10 km/h; Cascading alert triggered

Technical Architecture Summary

4.1 Shared Technology Stack

Library / Tool	Version	Usage Across Projects
Leaflet.js	1.9.4	Geospatial tile rendering, marker management, popup binding — PS1, PS3, PS4
Chart.js	4.4.1	Line, bar, and multi-series charts for analytics and LSTM prediction — PS3, PS4
HTML5 Canvas API	Native	Real-time CCTV simulation and YOLO bounding box overlay — PS4
HTML5 File API	Native	Drag-and-drop file ingestion (CSV, JSON, JPG) — PS1
CartoDB Tile Servers	CDN	Dark Matter tiles (PS1, PS4), Light All tiles (PS3) — terrain-appropriate base maps
Google Fonts CDN	CDN	Orbitron, Rajdhani, Share Tech Mono, Syne, Space Grotesk, IBM Plex Mono, Outfit

4.2 Deployment Architecture

- Platform: Vercel (serverless CDN deployment — zero backend infrastructure required)
- Build: Static HTML/CSS/JS — no build step, no bundler, no framework overhead
- Repository: github.com/ganeshkrishnareddy/Dashboards_Built_With_Python
- Access: Public, no authentication, no login wall — open directly from any URL
- Scalability: CDN-served; supports global concurrent access without server provisioning

4.3 Production Extensibility

All three dashboards are architected for zero-friction production upgrades:

Demo Layer	Production Replacement	Change Required
Simulated MongoDB ingest	MongoDB Atlas Data API	Replace ingestMongoDB() with fetch() call to /api/v1/action/find
Simulated S3 assets	AWS SDK v3 GetObjectCommand	Replace imageUrl with pre-signed S3 URL from SDK

Hardcoded GVS data	REST API / Google Sheets API	Replace CITIES object with async fetch from data endpoint
Canvas CCTV simulation	HLS/WebRTC stream + YOLO-v8 API	Replace canvas draw loop with <video> element + overlay canvas
Simulated LSTM curves	TensorFlow.js / Python API endpoint	Replace hardcoded curve data with fetch from ML inference API

Candidate Profile & Contact

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Project Repository

Dashboards_Built_With_Python

Live Deployments

Landing Page

PS1 — STRATFUSE Intelligence

PS3 — URBANVECTOR Growth

PS4 — TRAFFIQ Traffic Intel

Declaration

I hereby declare that all applications submitted are original work developed solely by me for the CYBERJOAR AI placement-drive assignment OC.41335.2026.59218. All live URLs are publicly accessible with no authentication required. The source code is available in the GitHub repository linked above.

P Ganesh Krishna Reddy · Reg. No. 12212186 · Lovely Professional University · 20 April 2026